Aquaculture Research

Aquaculture Research, 2016, **47**, 1631–1638 doi: 10.1111/are.12624

Effects of a dietary administration of purple coneflower (*Echinacea purpurea*) on growth, antioxidant activities and 8 miRNAs expressions in crucian carp (*Carassius auratus*)

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Abstract

Echinacea purpurea (EP), a globally popular herbal medicine, has been used to treat various diseases in human and animals. However, little has been reported about its effects in fish. In this study, crucian carp (*Carassius auratus*) were selected to evaluate the effects of EP on growth performance and antioxidant response and the expressions of microRNAs. The results showed EP could stimulate the growth of crucian carp with the best effect was observed at dose of 4 g kg^{-1}. In serum, the content of hydroxyl radical (·OH) and malondialdehyde (MDA) decreased by EP supplementation, whereas the activities of superoxide dismutase (SOD), catalase (CAT) and glutathione reductase (GR) increased. Similarly, the content of ·OH decreased, and the activities of CAT, GPX and GR increased in liver of crucian carp. Furthermore, in livers of crucian carp, EP supplements upregulated the expressions of the microRNAs (miR-16, miR-21, miR-125b, miR-146a, miR-155, miR-181a and miR-223), which had been confirmed to participate in regulating antioxidant and immune function in mammals. Our results suggest EP supplements in diets stimulated growth performance and antioxidant response of crucian carp. In liver, the upregulation of specific miRNAs may participate in the antioxidant effects of EP diets.

Keywords: *Echinacea purpurea*, growth, antioxidant activity, microRNA, crucian carp (*Carassius auratus*)

Introduction

Antibiotics and chemotherapeutics used for prophylaxis and treatment in aquaculture have been widely criticized for their side effects such as toxicity, resistance, residual impacts on public health and environmental hazards (FAO/WHO/OIE 2006). Recently, immunostimulants of herbal origin have been shown to possess the ability to increase disease resistance in fish by enhancing non-specific and specific defence mechanisms (Dügenci, Arda & Candan 2003; Harikrishnan, Balasundaram & Heo 2011; Tang, Cai, Liu, Wang, Lu, Wu & Jian 2014).

Echinacea purpurea (EP) is a globally popular herbal medicine used for many years by indigenous people of North America as a panacea for a variety of diseases including infections, trauma, inflammation and fever (Hobbs 1990). Pharmacological researches had shown the active ingredients of EP could stimulate T cells to promote phagocytosis as well as to enhance lymphocyte activity, which could improve the cellular oxidative status, anti-inflammatory and antibacterial activities etc. (Grimm & Müller 1999; Sun, Currier & Miller 1999; Weber, Taylor, Stoep, Weiss, Standish & Calabrese 2005; Matthias, Banbury, Bone, Leach & Lehmann 2008). EP also showed effects on growth promotion and immunoregulation in fish culture (Aly, Mohamed & John 2008; Aly & Mohamed 2010; Oskoii, Kohyani, Parseh, Salati & Sadeghi 2012).

MicroRNAs (miRNAs) are endogenous non-protein-coding RNAs with ~22 nucleotides in length. They exert down-regulation ability, either by translation inhibition or by degradation mRNA, on target genes through complementary binding to their 3′-UTR regions (Bartel 2004). Recent studies showed that miRNAs might have potentially enormous importance in the regulation of many fundamental biological processes, such as immunological and inflammatory disorders (Lindsay 2008; Sonkoly, Stahle & Pivarcsi 2008a,b). Over the past few years, the researches on miRNAs have only been carried in several species such as zebrafish, medaka, the Asian seabass and channel catfish (Kloosterman, Steiner, Berezikov, de Bruijn, van de Belt, Verheul, Cuppen & Plasterk 2006; Tani, Kusakabe, Naruse, Sakamoto & Inoue 2010; Xia, He, Bai & Yue 2011; Xu, Chen, Li, Ge, Pan & Xu 2013). Although many endogenous miRNAs have been identified in fish, the specific functions have remained largely undefined until now. To date, no miRNAs are available for the crucian carp (*Carassius auratus*).

Researches in mammals indicated miRNAs have significant relationships with oxidative stress. The expressions of some miRNAs were significantly altered by oxidative stress (Bai, Ma, Ding, Fu, Shi & Chen 2011; Muratsu-Ikeda, Nangaku, Ikeda, Tanaka, Wada & Inagi 2012). Mateescu, Batista, Cardon, Gruosso, de Feraudy, Mariani, Nicolas, Meyniel, Cottu, Sastre-Garau and Mechta-Grigoriou (2011) showed a new mechanism of action for miR-141 and miR-200a in oxidative stress response by identifying p38α MAPK as one of their relevant targets. On the other hand, miRNAs have also been confirmed to regulate many aspects of the immune functions in mammals by regulating T- and B-cell differentiation (Chen, Li, Lodish & Bartel 2004), myeloid production (Fazi, Rosa, Fatica, Gelmetti, De Marchis, Nervi & Bozzoni 2005; Johnnidis, Harris, Wheeler, Stehling-Sun, Lam, Kirak, Brummelkamp, Fleming & Camargo 2008), the acute inflammatory response after the recognition of pathogens (Taganov, Boldin, Chang & Baltimore, 2006; O'Connell, Taganov, Boldin, Cheng & Baltimore 2007; Tili, Michaille, Cimino, Costinean, Dumitru, Adair, Fabbri, Alder, Liu, Calin & Croce 2007), and B- and T-cell responses (O'Connell *et al.* 2007; Tili *et al.* 2007).

Studies have shown that most of the miRNA sequences and the stems of miRNA hairpins were highly evolutionary conserved across species (Berezikov, van Tetering, Verheul, van de Belt, van Laake, Vos, Verloop, van de Wetering, Guryev, Takada, van Zonneveld, Mano, Plasterk & Cuppen 2006). Therefore, miRNAs could be deduced to participate in regulating the antioxidant and immune functions of fish though there were few studies about that till now. In this study, we evaluated the effects of EP on growth performance and antioxidant activity of crucian carp firstly. To uncover the possible mechanisms involved in EP's effects in fish, livers were chosen to detect the expressions of 8 miRNAs (miR-16, miR-21, miR-122, miR-125b, miR-146a, miR-155, miR-181a and miR-223) for liver is one of the most important organs in response to the oxidative stress and pathogens infection (Sturve, Almroth & Förlin 2008; Sinha, AbdElgawad, Giblen, Zinta, De Rop, Asard, Blust & De Boeck 2014). This study supplied the basis for the use of EP and help understand the functions of miRNAs in fish.

Material and methods

Experimental diets

Formulation and proximate composition of the test diets were shown in Table 1. Diets supplemented with four levels of EP (0, 1, 2 and 4 g kg^{-1}) were prepared by adding EP extract (4% chicoric acid grade, from Xi'an Tianyi Biotechnology, Xi'an, China), which were assigned to control, EP1, EP2 and EP4 group respectively. Ingredients were weighed and mixed thoroughly, followed by addition of oil and water. The wet mash was pelleted with a 1.0-mm in diameter. The resulting pellets were air-dried and then stored at −20°C until used.

Animal rearing

Crucian carp (*Carassius auratus* Pengze) were obtained from Beijiang Institute of Aquatic Sciences (Qingyuan, Guangdong, China). Fish were stocked in a quarantine tank for 15 days to assess their health status. After that, fish (mean individual initial weight 7.50 ± 0.15 g) were randomly distributed in four equal groups. Each group consisted of three replicates of 30 fish per replicate, reared in indoor recirculating aquaculture system (each, 1 m^3) and fed with various diets for 60 days. The daily feeding rate was 4–5% body weight and was adjusted according to prior feeding responses. The temperature and pH of water were 26.6 ± 1.2°C and 7.4–7.8, respectively,

Table 1 Formulation and nutrient composition of experimental diets

	Content in each group (%)			
Ingredients (%)	**Control**	**EP1**	**EP2**	**EP4**
Fish meal	18	18	18	18
Soybean meal	19	19	19	19
Rapeseed meal	23	23	23	23
Cottonseed meal	8	8	8	8
Flour	25.1	25	24.9	24.7
Soybean oil	4	4	4	4
Choline chloride	0.2	0.2	0.2	0.2
Calcium dihydrogen phosphate	1.5	1.5	1.5	1.5
Sodium chloride	0.2	0.2	0.2	0.2
Vitamin mix*	0.5	0.5	0.5	0.5
Mineral mix†	0.5	0.5	0.5	0.5
Echinacea purpurea	0	0.10	0.20	0.40
Proximate composition (%)				
Crude protein	30.27	30.18	30.25	30.47
Crude lipid	6.58	6.46	6.82	6.56
Ash	10.83	10.65	10.78	10.69

*Vitamin mix (Xinjian Feed Company, Guangdong, China). (Kg^{-1}): Vitamin A, 2 mg; Vitamin B1, 20 mg; Vitamin B2, 20 mg; Vitamin B6, 10 mg; nicotinic acid, 100 mg; Pantothenic acid, 50 mg; Biotin, 1 mg; Folic acid, 5 mg; Inositol, 500 mg; Vitamin E, 50 mg; Vitamin B12, 0.02 mg; Vitamin K3, 10 mg; and Vitamin D3, 0.05 mg.

†Mineral mix (Xinjian Feed Company, Guangdong, China). (Kg^{-1}): $ZnSO_4 \cdot 7H_2O$, 525.46 mg; $MnSO_4 \cdot H_2O$, 49.22 mg; KI, 5.23 mg; $FeSO_4 \cdot 7H_2O$, 238.83 mg; $MgSO_4 \cdot 7H2O$, 1.6 g; $CuSO_4 \cdot 5H_2O$, 11.82 mg; $CoCl_2 \cdot 6H_2O$, 0.2 mg; Na_2SeO_4, 0.66 mg; KCl, 600 mg; and NaCl, 107.08 mg.

during the feeding period. Dissolved oxygen was not less than 5.0 mg L^{-1} and concentrations of ammonia-N and nitrite-N were less than 0.1 mg L^{-1}.

Growth performance and survival

The growth performance and survival parameters were calculated at the end of the experiments. Fish of all replicates were weighed individually, and the indexes were calculated by the following formulas (Oskoii *et al.* 2012).

Weight gain rate (WGR, %) = 100 × (final body weight − initial body weight)/initial body weight

Feed efficiency (FE, %) =100 × (weight gain/total feed intake)

Hepatosomatic index (HSI, %) = 100 × (liver weight)/(body weight)

Survival rate (SR, %) = 100 × (final number of fish)/(initial number of fish)

Antioxidant activity analysis

At the end of the experiment, body weights of fish were measured. The serums and livers were sampled and stored at −80°C until used. The activities of superoxide dismutase (SOD), catalase (CAT), glutathione peroxidase (GPX), glutathione reductase (GR) and the contents of hydroxyl radical (·OH) and malondialdehyde (MDA) were determined according to the method described previously (Liu, Xi, Yang, Li, Jiang, Shu, Wang, Gao, Zhu & Zhang 2011).

Expressions of microRNAs analysis in liver by quantitative real-time PCR

At the end of the experiment, the livers of fish from each group were collected and all samples were frozen in liquid nitrogen and stored at −80°C until total RNA isolation. Total RNA was isolated from the samples using the Trizol (Invitrogen, Carlsbad, CA, USA) reagent according to the manufacturer's protocol (Zhong, Holzgreve & Huang 2008). The quality of RNA was examined by 1.5% agarose gel electrophoresis and with BioPhotometer 6131 (Eppendorf, Hamburg, Germany).

Stem-loop RT-PCR was performed as previously described (Chen, Ridzon, Broomer, Zhou, Lee, Nguyen, Barbisin, Xu, Mahuvakar, Andersen, Lao, Livak & Guegler 2005). In brief, 1 μg of total RNA from each sample was reverse-transcribed into cDNA by the M-MLV reverse transcriptase (Promega, Guanzhou, China) with looped antisense primers. After 1 h of incubation at 42°C and 10 min of deactivation at 85°C, the reaction mixes were used as the templates for PCR. Real-time quantitative PCR was performed with standard protocols on an ABI PRISM® 7500 sequence detection system (Applied Biosystems, Foster, CA, USA). The PCR mixture contained 1 μL of cDNA (1:10 dilution of a RT-reaction mix), 10 μL of 26 SYBR Green PCR Master Mix (Toyobo, Osaka, Japan), 1.5 μM of each primer, and water to make up the final volume to 20 μL. The reaction was performed in a 96-well optical plate at 95°C for 5 min, followed by 40 cycles of 95°C for 15 s, 65°C for 15 s and 72°C for 32 s. All reactions were run in duplicates, and a negative control without template was included for each gene. The cycle threshold (Ct) was recorded for each reaction and the amount of each miRNA relative to that of

U6 RNA was described using the expression $2^{-(\mathrm{CtmiRNA\text{-}CtU6RNA})}$. Primers were designed on the basis of the sequenced miRNA by using Premier 5.0.

Statistics

Data were expressed as means ± SE (standard error). Means were analyzed by one-way analysis of variance (ANOVA) test using SPSS 17.0. Significant difference and means were tested using Duncan's test to compare the means of each groups. $P < 0.05$ was the accepted significance level.

Results

Effects of EP diets on growth performance and survival of crucian carp

As shown in Table 2, EP increased the growth of crucian carp. The weight gain rates in EP4 group increased significantly compared with the control and other experimental groups ($P < 0.05$). *Echinacea purpurea* had no significant effects on FE, HIS and SR of crucian carp ($P > 0.05$).

Effects of EP diets on antioxidant activities in crucian carp

Effects of EP diets on antioxidant activities in serums of crucian carp revealed in Table 3 that dietary treatment EP4 (4 g kg^{-1}) significantly decreased the content of ·OH and MDA, increased SOD, CAT and GR activities and decreased GPX activities ($P < 0.05$).

Table 4 showed the effects of EP diets on antioxidant activities in livers of crucian carp. *Echinacea purpurea* significantly decreased the content of ·OH, increased CAT, GPX and GR activities ($P < 0.05$), however, there were no significant effects on SOD activity and the content of MDA ($P > 0.05$).

Effects of EP diets on expressions of 8 miRNAs in livers of crucian carp

The expressions of 8 miRNAs in livers of the crucian carp with EP supplementation were investigated (Fig. 1). *Echinacea purpurea* significantly affected the expressions of most miRNAs dectected in present study. The expressions of miR-16, miR-21, miR-146a and miR-155 were significantly increased in the EP2 and EP4 group ($P < 0.05$), and the up-regulations of miR-125b, miR-181a and miR-223 were observed in the EP4 group ($P < 0.05$). Of all the miRNAs, miR-223 had the highest change affected by EP4 (30 fold) followed by miR-155 (12 fold).

Discussion

Echinacea purpurea has been widely used in many fields (Sun *et al.* 1999; Weber *et al.* 2005; Matthias *et al.* 2008). In this study, EP supplements have the best effect on growth performance in crucian carp at the dose of 4 g kg^{-1}. Similar stimulatory effects on WGR were observed when EP diets were fed Nile Tilapia (*Oreochromis niloticus*) (Aly *et al.* 2008; Aly & Mohamed 2010). In addition, EP-added diets also showed increase in feed conversion ratio (FCR) and SGR in guppy (*Poecilia reticulate*) (Guz, Sopinska & Oniszczuk 2011) and increase of WG, FCR and SGR in rainbow trout (*Oncorhynchus mykiss*) (Oskoii *et al.* 2012).

Antioxidant activity is one of the defensive abilities in body and the defensive system comprises of antioxidant enzymes, such as SOD, CAT GPX and GR in fish (Sturve *et al.* 2008; Tang *et al.* 2014). The antioxidant property of EP has been demonstrated previously (Chen, Hu, Raghubeer & Kitts

Table 2 Effects of EP-supplemented diets on the growth performance of crucian carp

Treatment	WGR (%)	FE (%)	HSI (%)	SR (%)
Control	162.60 ± 10.30[b]	43.95 ± 3.52	4.28 ± 0.38	97.78 ± 1.11
EP1	182.90 ± 12.93[ab]	44.68 ± 3.00	4.44 ± 0.23	94.44 ± 2.22
EP2	184.09 ± 8.76[ab]	45.08 ± 1.94	4.70 ± 0.16	96.67 ± 1.92
EP4	212.32 ± 1.82[a]	49.35 ± 2.33	4.73 ± 0.04	95.56 ± 4.44

[a, b] Means in the same column, not sharing a common superscript letter were significantly different ($P < 0.05$) as determined by Duncan's test. Values are means ± SE, $n = 3$. EP, *Echinacea purpurea*.
FE, feed efficiency; HSI, hepatosomatic index; SR, survival rate; WGR, weight gain rate.

Table 3 Effects of EP-supplemented diets on antioxidant activity in serums of crucian carp

Treatment	·OH (U mL^{-1})	SOD (U mL^{-1})	CAT (U mL^{-1})	GPX (U mL^{-1})	GR (U L^{-1})	MDA (nmol mL^{-1})
Control	1925.59 ± 12.57^{a}	81.83 ± 1.88^{b}	35.32 ± 0.54^{c}	257.10 ± 2.90^{a}	10.85 ± 0.10^{b}	125.00 ± 8.87^{a}
EP1	1489.78 ± 80.98^{b}	88.58 ± 2.07^{ab}	41.56 ± 1.09^{b}	249.00 ± 1.15^{a}	12.28 ± 0.55^{ab}	112.14 ± 3.89^{a}
EP2	1445.43 ± 21.43^{b}	92.68 ± 2.60^{a}	46.07 ± 0.99^{ab}	213.30 ± 7.70^{b}	12.76 ± 0.83^{ab}	111.32 ± 0.41^{a}
EP4	1482.34 ± 79.75^{b}	95.56 ± 2.37^{a}	50.82 ± 2.22^{a}	210.30 ± 5.30^{b}	14.76 ± 0.28^{a}	93.57 ± 2.78^{b}

[a, b, c] Means in the same column, not sharing a common superscript letter were significantly different ($P < 0.05$) as determined by Duncan's test. Values are means ± SE, $n = 3$. EP, *Echinacea purpurea*.

CAT, catalase; GPX, glutathione peroxidase; GR, glutathione reductase; MDA, malondialdehyde; ·OH, hydroxyl radical; and SOD, superoxide dismutase.

Table 4 Effects of EP-supplemented diets on antioxidant activity in livers of crucian carp

Treatment	·OH (U mg^{-1} pr)	SOD (U mg^{-1} pr)	CAT (U mg^{-1} pr)	GPX (U mg^{-1} pr)	GR (U g^{-1} pr)	MDA (nmol mg^{-1} pr)
Control	5553.64 ± 26.18^{a}	7.08 ± 0.56	27.56 ± 1.48^{b}	683.11 ± 12.67^{b}	135.58 ± 7.04^{b}	5.74 ± 0.39
EP1	3040.83 ± 89.71^{d}	7.81 ± 0.32	30.79 ± 3.72^{b}	622.55 ± 12.42^{b}	114.14 ± 4.65^{b}	5.19 ± 0.40
EP2	3977.47 ± 231.48^{c}	6.58 ± 0.17	52.41 ± 4.54^{a}	774.61 ± 34.28^{b}	167.25 ± 17.30^{a}	5.36 ± 0.14
EP4	4845.86 ± 148.23^{b}	6.43 ± 0.52	26.59 ± 2.61^{b}	1181.10 ± 71.56^{a}	195.19 ± 7.18^{a}	4.87 ± 0.39

[a, b, c, d] Means in the same column, not sharing a common superscript letter were significantly different ($P < 0.05$) as determined by Duncan's test. Values are means ± SE, $n = 3$. EP, *Echinacea purpurea*.

CAT, catalase; GPX, glutathione peroxidase; GR, glutathione reductase; MDA, malondialdehyde; ·OH, hydroxyl radical; and SOD, superoxide dismutase.

2010; Lee, Ciou, Chen & Yu 2012). In this study, the increased activities of SOD, CAT and GR, and the decreased content of ·OH and MDA in serums may indicate the antioxidant abilities of EP on the whole body. EP also affected the content of ·OH and the activities of CAT, GPX and GR in liver of crucian carp in this study.

Growing evidence has demonstrated that the miRNAs involved in the oxidative stress. MiR-21 participates in H_2O_2-mediated gene regulation and cellular injury response through programmed cell death (Cheng, Liu, Zhang, Lin, Yang & Zhang 2009). Expressions of miR-21, miR-155 and miR-205 was significantly altered by oxidative stress and miR-205 regulated intracellular ROS levels via the alteration of anti-oxidant enzymes, such as SOD1, SOD2, CAT, and HO-1 (Muratsu-Ikeda *et al.* 2012). In this study, the expressions of miR-21 and miR-155 were up-regulated in liver, which might result in the decreased content of ·OH and increased activities of CAT and GR affected by EP. MiR-125b was also up-regulated by EP, and similar effects were observed by Gaedicke, Zhang, Schmelzer, Lou, Doering, Frank and Rimbach (2008) who proved vitamin E, a typical antioxidant, resulted in increased concentrations of miR-125b in rat liver. GPX was the potential target gene of miR-21 and miR-223 (Bai *et al.* 2011). Up-regulation of miR-223 and miR-21 were observed to be coinstantaneous with the increase in GPX activity in liver with supplementation of EP. The induction of miR-223 and miR-21 expressions in fish might also have occurred by a GPX-independent mechanism. Therefore, the expressions of miRNAs were affected by EP may indicate miRNAs involved in regulating the antioxidant activity, however, the roles of miRNAs in regulating the target genes of anti-oxidation still need further investigations.

As a globally popular herbal immuno-modulator, the immunoregulating effects of EP have been proved by many researches in human, in other animals and in fish (Currier, Sicotte & Miller 2001; Zhai, Solco, Wu, Wurtele, Kohut, Murphy & Cunnick 2009; Aly & Mohamed 2010; Hudson 2012; Oskoii *et al.* 2012). MiRNAs have been shown to regulate many aspects of the immune functions in mammals (Costinean, Zanesi, Pekarsky, Tili, Volinia, Heerema & Croce 2006; O'Connell *et al.* 2007; Tili *et al.* 2007; Johnnidis *et al.* 2008). In crucian carp, the expressions of the miRNAs were up-regulated by EP might

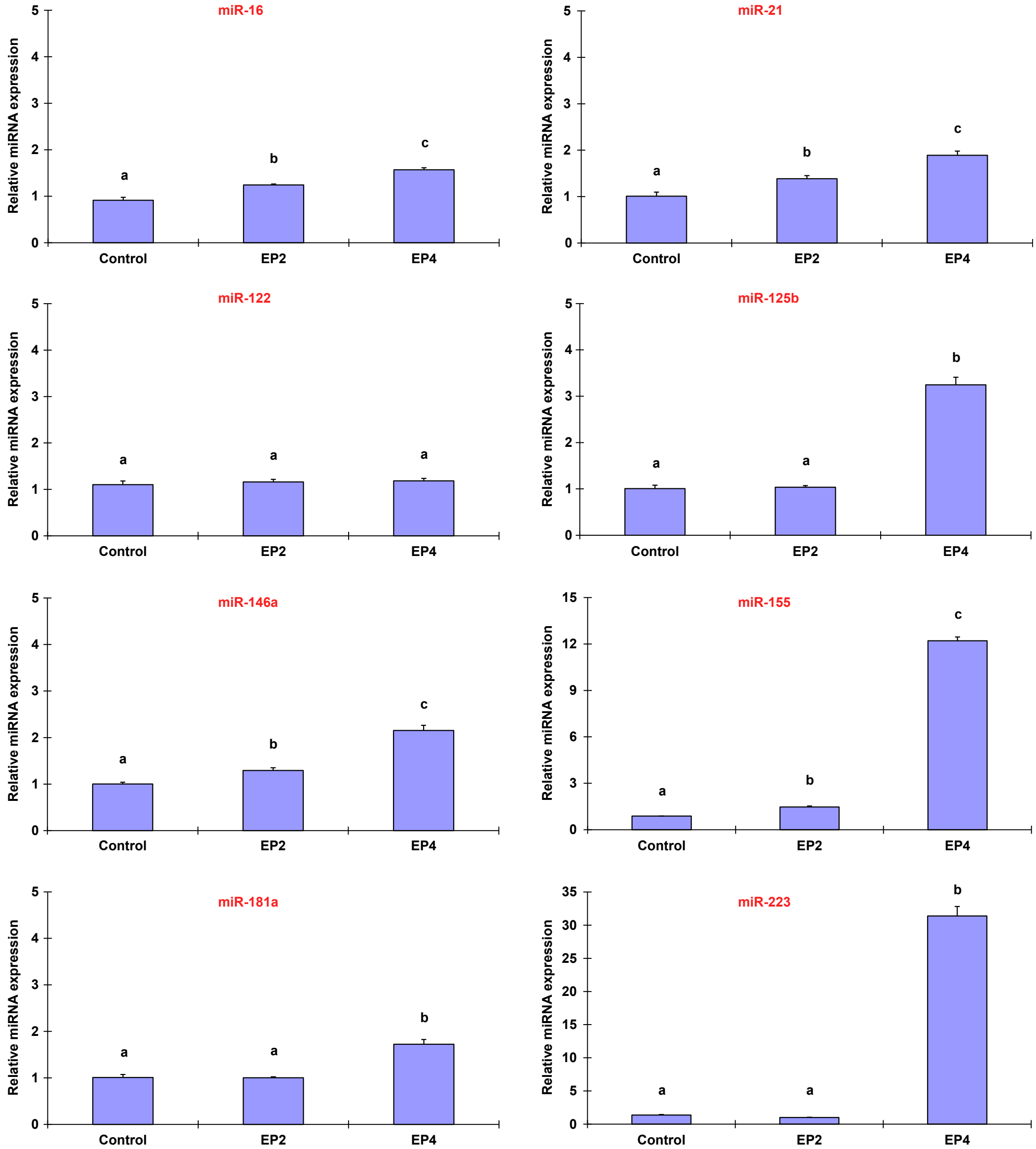

Figure 1 Effects of EP diets on expressions of 8 microRNAs in liver of crucian carp. At the end of the experiments, the livers of fish from each group were collected and all samples were frozen in liquid nitrogen and stored at −80°C until total RNA isolation. Total RNA was isolated from the samples using the Trizol. The expressions of 8 microRNAs were analyzed by quantitative real-time PCR. The letters a, b, c mean, in the same column, not sharing a common superscript letter was significantly different ($P < 0.05$) as determined by Duncan's test. Values are means ± SE, $n = 3$. EP, *Echinacea purpurea*.

suggested miRNAs participate in regulating immune function in fish. Similar findings were also reported by Xia *et al.* (2011) that exposure of the Asian seabass (*Lates calcarifer*) to lipopolysaccharide resulted in up-regulation of over 50% of the identified miRNAs in spleen including the miR-21, miR-125 and miR-181 etc. The miR-21 was further detected to be highly up-regulated at 24 h post challenge by *Vibrio harveyi* in liver of seabass (Xia *et al.* 2011). The above results suggested the importance of the miRNAs in immune responses of fish, though there were only few reports clarifying this function in fish.

In summary, this study clearly demonstrated for the first time that EP-supplemented diets increased the growth performance and antioxidant responses of crucian carp. How EP affected the antioxidant and immunoregulation in fish might partly be explained by EP regulating the expressions of 8 antioxidant- or immune-related miRNAs, while the further mechanisms still need more investigations.

Acknowledgments

This work was supported by grants from the National Natural Science Foundation of China (31302213 and 31001117), the earmarked fund for China Agriculture Research System (CARS-46-43), and the Project on the Integration of Industry, Education and Research of Guangdong Province and the Education Ministry of China (2012B091100421).

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